Sustainable Development Goal Readiness Index 2030

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1. Introduction

1.1 Project Aims

This project aims to develop a composite indicator that measures how prepared countries are to achieve selected Sustainable Development Goals (SDGs) by 2030. The project focuses on three specific SDGs:

- SDG 3 - Good Health and Well-being
- SDG 6 - Clean Water and Sanitation
- SDG 11 - Sustainable Cities and Communities

To achieve this, multiple international datasets were collected and combined to create separate sub-indexes for each SDG. These sub-indexes were then aggregated into a final score known as the SDG Readiness Index 2030.

Another aim of the project is to apply a range of data pre-processing and statistical techniques throughout the development process. This includes handling of missing data, detecting outliers, normalising indicators, forecasting future values for 2030 and combining the indicators into a single composite indicator.

The project also aims to move beyond historical analysis by estimating future SDG readiness levels per country. By forecasting indicator values towards 2030, this attempts to provide insight to how prepared countries may be in the future rather than only evaluating a country's past performance.

In addition, multivariate analysis techniques such as Principal Component Analysis (PCA) and K-Means clustering were used to further analyse the relationships and patterns between the SDG sub-indexes. These methods helped identify similarities, differences and grouping patterns among countries within the final dataset.

The final aim is to present the findings through clear visualisations and rankings so that differences between countries can be easily interpreted and compared.

1.2 Limitations of this project

One limitation of this project is the availability and quality of data. Some countries contained missing values and incomplete data across certain indicators and years. As a result, some data required imputation, while some countries were excluded during dataset merging where matching country data was unavailable.

Another limitation is that some datasets only contained more recent years of data, such as 2021, 2024 or 2025. This reduced the amount of historical information available for certain indicators. However, these datasets were still included in the project because they provided enough trend-pattern information to support forecasting and overall readiness analysis.

Another limitation is that the project only focuses on three SDGs rather than all 17 goals established by the United Nations. Although SDG 3, SDG 6 and SDG 11 provide meaningful insight into health, sanitation and sustainability, the final index does not represent overall SDG performance in its entirety.

The 2030 predictions are based on historical trends, meaning the forecasts assume that previous patterns will continue into the future. Unexpected factors such as economic instability, government policy changes, conflicts, environmental disasters or even global crises may affect future outcomes differently.

2. Theoretical Framework

2.1 Composite Indicator Theory

Composite indicators are used to combine multiple individual indicators into a single measurable score. They are commonly used to measure complex topics such as sustainability, economic development and quality of life, where one variable alone cannot fully represent the overall concept.

According to the Organisation for Economic Co-operation and Development Handbook on Constructing Composite Indicators:

"A composite indicator is formed when individual indicators are compiled into a single index on the basis of an underlying model."

This theory forms the foundation of this project as well. Since SDG readiness cannot be measured using a single indicator, multiple datasets related to SDG 3, SDG 6 and SDG 11 were combined into separate sub-indexes before being aggregated into a final Global SDG Readiness Index 2030.

The methodology used throughout this project was heavily influenced by the OECD framework, particularly in areas such as data selection, missing data handling, normalisation, weighting, aggregation and visualisation.

2.2 SDG Measurement Concepts

The United Nations Sustainable Development Goals (SDGs) are a collection of 17 global goals designed to address major social, economic and environmental challenges by 2030. Each SDG is measured using specific indicators that allow countries to monitor progress over time.

This project focuses on the goals for SDG 3, SDG 6 and SDG 11 by using selected indicators related to health, sanitation and sustainable cities. Since each indicator measures different units and scales, techniques such as normalisation and aggregation are required to make the data comparable across countries.

The use of sub-indexes also allows related indicators to be grouped together under each SDG before contributing to the final composite readiness score.

2.3 Why SDG Readiness Matters

Measuring SDG readiness is important because it helps identify how prepared countries are to achieve sustainable development goals by 2030. Readiness analysis can highlight the strengths, weaknesses and development gaps between countries. This then allows policymakers and organisations to better understand where improvement is needed.

Rather than only analysing past performance, readiness-focused approaches attempt to estimate future preparedness using current and historical trends. This provides a more forward-looking view of global development progress.

By creating a composite SDG readiness index, this project aims to simplify complex datasets into a more understandable format that supports comparison, interpretation and decision-making.

3. Data Selection

3.1 Data Collection

There was a total of 8 datasets used in this project, which were collected from the official United Nations SDG Data Portal. The project focuses on three Sustainable Development Goals:

- SDG 3 — Good Health and Well-Being (SH_DYN_IMRT, SH_DTH_NCOM, SH_STA_SCIDE)
- SDG 6 — Clean Water and Sanitation (ER_H2O_STRESS, SH_H2O_SAFE, SH_SAN_HNDWSH)
- SDG 11 — Sustainable Cities and Communities (slum_df, disaster_df)

Multiple indicators were selected under each SDG category to provide measurable insight into health, sanitation and sustainability performance across countries.

The datasets obtained from the UN SDG database were already internationally standardised and structured. As a result, the data provided consistent country-based measurements and indicator formats across different years and regions. This improved the reliability and comparability of the datasets used within the project.

Although the datasets were standardised by the UN, some indicators still contained missing values or incomplete yearly records for certain countries. These limitations were taken into consideration during the overall development of the composite indicator.

See 4. Imputation of Missing Data

3.2 Indicator Justification

1. SDG 3: Good Health and Well-Being

Three indicator datasets were selected for SDG 3 to measure different areas of public health and well-being.

The infant mortality rate (SH_DYN_IMRT) was chosen because it reflects healthcare quality, maternal care and living conditions within a country.

The mortality rate from cardiovascular disease, cancer, diabetes and chronic respiratory disease (SH_DTH_NCOM) was selected because non-communicable diseases are among the leading causes of death globally and represent long-term population health.

The suicide mortality rate (SH_STA_SCIDE) was included to represent mental health and social well-being.

Together, these indicators provide a balanced view of healthcare, chronic illness and mental health performance.

Dataset	GeoAreaName	Sex	2000	2001	....
SH_DYN_IMRT	Afghanistan	BOTHSEX	110.60	107.51	....
SH_DTH_NCOM	Afghanistan	BOTHSEX	43.2	41.9	....
SH_STA_SCIDE	Afghanistan	BOTHSEX	4.4	4.2	....

An example of what a row per dataset would look like.

2. SDG 6: Clean Water and Sanitation

Three indicator datasets were selected for SDG 6 to measure water sustainability and sanitation access.

The level of water stress (ER_H2O_STRESS) was chosen because it measures pressure on freshwater resources and long-term water sustainability.

The proportion of the population using safely managed drinking water services (SH_H2O_SAFE) was selected because access to safe drinking water is a core target of SDG 6.

The proportion of the population with basic handwashing facilities (SH_SAN_HNDWSH) was included because sanitation and hygiene are directly linked to public health and disease prevention.

Together, these indicators measure water availability, safe access and sanitation quality.

Dataset	GeoAreaName	Observation	2000	2001	....
ER_H2O_STRESS	Afghanistan	E	0.35	0.37	....
SH_H2O_SAFE	Afghanistan	G	11.48	11.49	....
SH_SAN_HNDWSH	Afghanistan	G	....	....	....

An example of what a row per dataset would look like.

3. SDG 11: Sustainable Cities and Communities

Two indicators were selected for SDG 11 to measure urban living conditions and disaster vulnerability.

The proportion of the urban population living in slums (EN_LND_SLUM) was chosen because it reflects overcrowding, poor housing conditions and limited urban infrastructure.

The number of directly affected persons attributed to disasters per 100,000 population (VC_DSR_DAFF) was included because it measures the impact of disasters and a country's resilience to environmental risks.

Together, these indicators provide insight into urban sustainability, safety and resilience.

Dataset	GeoAreaName	2005	2006	....
EN_LND_SLUM	Afghanistan	63.6	67.1	....
VC_DSR_DAFF	Afghanistan	0.35	0.09	....

An example of what a row per dataset would look like.

After justifying the indicators per dataset, they were then translated into a dataframe, which follows:

Dataset Name	Converted Dataframe
SH_DYN_IMRT	infant_df
SH_DTH_NCOM	disease_df
SH_STA_SCIDE	suicide_df
ER_H2O_STRESS	stress_df
SH_H2O_SAFE	drinking_df
SH_SAN_HNDWSH	handwash_df
EN_LND_SLUM	slum_df
VC_DSR_DAFF	disaster_df

3.3 Time-Driven Condition

A time-driven condition was applied throughout the project to maintain consistency across all datasets and indicators. The general rule was to select data from the year 2000 up to the most recent year available within each dataset. This approach ensured that the project used long-term historical trends while still incorporating the latest available information.

However, some indicators did not contain complete yearly records from 2000 onwards. In particular, certain datasets such as the slums population and handwash facilities indicators were only available in wider interval periods rather than annual observations. Because of this, a 5-year interval structure was applied across the affected SDG indicators to maintain consistency within the forecasting process.

As a result, datasets were commonly structured using years such as: 2005, 2010, 2015, 2020and the most recent available year.

This approach allowed the project to preserve comparable trend patterns across indicators while still including the most up-to-date data available from the United Nations SDG database.

4. Imputation of Missing Data

Before merging the datasets into their respective SDG sub-indexes, missing data handling was first carried out individually on each dataset. This approach helped maintain consistency and prevented missing values from affecting the dataset merging for the composite indicator in the project.

4.1 Dealing with Duplicates

1. SDG 3 Datasets (infant_df, disease_df, suicide_df)

During the inspection of the SDG 3 datasets, duplicate country entries were identified across all three indicators. These duplicates were not treated as data errors because the datasets were separated by biological sex categories.

Each country appeared under: MALE, FEMALE and BOTHSEX

The BOTHSEX category represented the combined population measurement across both male and female groups. As a result, countries appeared multiple times within the datasets, but each row represented a different demographic grouping rather than a true duplicate observation.

To maintain consistency across the project, the combined BOTHSEX category was selected for analysis since it provided an overall country-level health measurement without separating the population by sex.

2. SDG 6 Dataset (stress_df, drinking_df, handwash_df)

During the inspection of the stress_df dataset, multiple rows were identified for the same country and year. These were not considered true duplicates because the dataset was divided across both activity categories and observation statuses.

Countries appeared under several activity categories, including:

- INDUSTRIES
- ISIC4_GTT
- ISIC4_A01_A0210_A0322
- TOTAL

These categories represented freshwater stress measurements across different economic sectors. Since the aim of the project was to measure overall national water stress rather than sector-specific stress, the TOTAL category was selected because it represented the combined freshwater withdrawal across all sectors.

The dataset also contained multiple observation statuses such as: A, E, I and O.

These statuses indicated whether the data was reported directly, estimated, imputed or obtained from different statistical bodies. Because some statuses contained incomplete records or inconsistent coverage, preference was given to the most complete and reliable observations during the cleaning process.

Within both the drinking_df and handwash_df datasets, countries also appeared multiple times per year. These were not treated as duplicate errors because the indicators were separated by location categories.

The datasets contained:

- ALLAREA
- RURAL
- URBAN

Each category represented a different population grouping based on location type. Since the project aimed to create comparable country-level readiness measurements, the ALLAREA category was selected because it represented the combined national measurement across both urban and rural populations.

4.2 Identifying Missing Data

1. SDG 6 Dataset (stress_df, drinking_df, handwash_df)

Within the stress_df dataset, missing values were commonly linked to different observation statuses. In many cases, a country and year contained missing values under one reporting source while valid values existed under another. After selecting only the TOTAL activity category, the remaining dataset was inspected to identify countries with excessive yearly gaps between 2000 and 2023.

A similar inspection process was applied to both drinking_df and handwash_df. Most missing observations appeared as isolated yearly gaps rather than complete missing sections across countries. This indicated that many countries still contained enough historical trend information to remain useful within the analysis.

2. SDG 11 Datasets (slums_df)

For the slum_df dataset, the missing data pattern was examined more closely due to the higher number of incomplete yearly observations. Many of the missing values appeared between existing years, suggesting that the overall country trends could still be reasonably estimated. However, countries containing very large sections of missing data were identified as unreliable for trend analysis and later removed from the dataset.

Overall, the missing data inspection stage focused on identifying:

- Countries with excessive missing observations.
- Isolated missing years.
- Consecutive missing periods.
- Whether enough historical trend information existed to support a realistic estimation.

4.3 Imputing Data

1. SDG 6 Dataset (stress_df, drinking_df, handwash_df)

For the stress_df, drinking_df and handwash_df datasets, countries with more than 10 missing yearly observations between 2000 and 2023 were removed before imputation began. Remaining missing values were estimated using neighbouring yearly observations together with each country's overall historical mean.

Where surrounding years were available, the missing value was estimated using the average of:

- The previous available year.
- The next available year.
- The country's overall average value.

This approach helped preserve the general increasing or decreasing trend within the data while keeping the estimated value aligned with the country's historical behaviour.

2. SDG 11 Datasets (slums_df)

For the slum_df dataset, missing values between existing yearly observations were estimated using interpolation techniques. This allowed the imputed values to follow the overall direction and rate of change already present within the country's trend pattern. Countries containing more than 20 missing observations were removed because the gaps were considered too large to estimate reliably.

Overall, the imputation process aimed to maintain realistic country-level trends while reducing the impact of incomplete records on later forecasting and composite indicator analysis.

4.4 Detecting Outliers

1. SDG 3 Datasets (infant_df, disease_df)

The infant_df dataset contained several extremely high values, mainly from countries such as Afghanistan and multiple countries within Sub-Saharan Africa. These values reflected genuine health challenges linked to poverty, conflict, poor healthcare access, malnutrition and disease rather than data errors. The outliers also appeared consistently across multiple years, showing long-term patterns rather than random spikes. However, the dataset was heavily right-skewed, meaning a small number of countries could disproportionately influence the analysis. To reduce this effect while still preserving the observations, values above the 95th percentile were capped for each year.

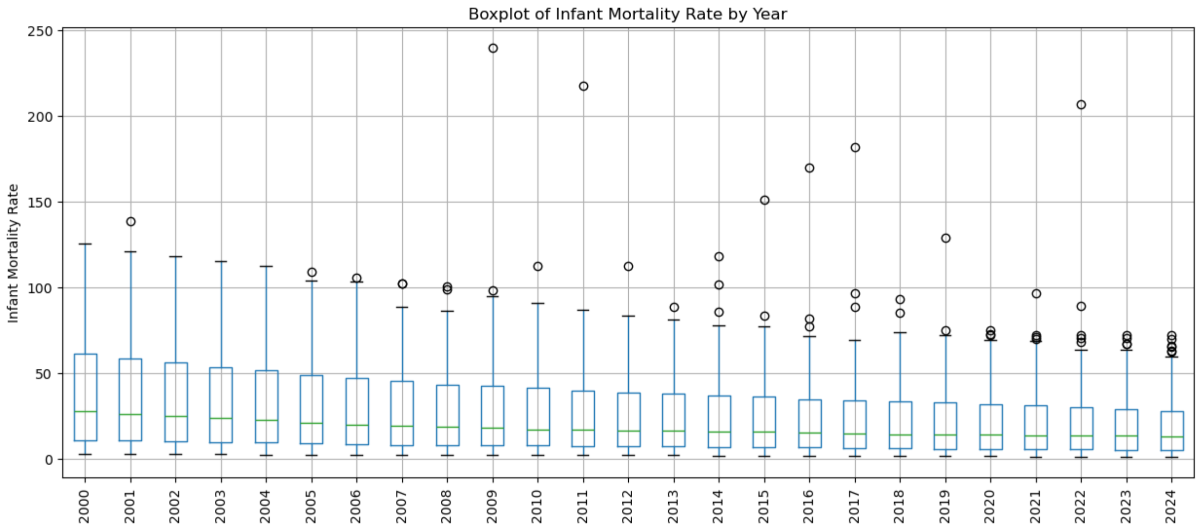

Figure 1. Infant Mortality Rate Before Outlier Capping

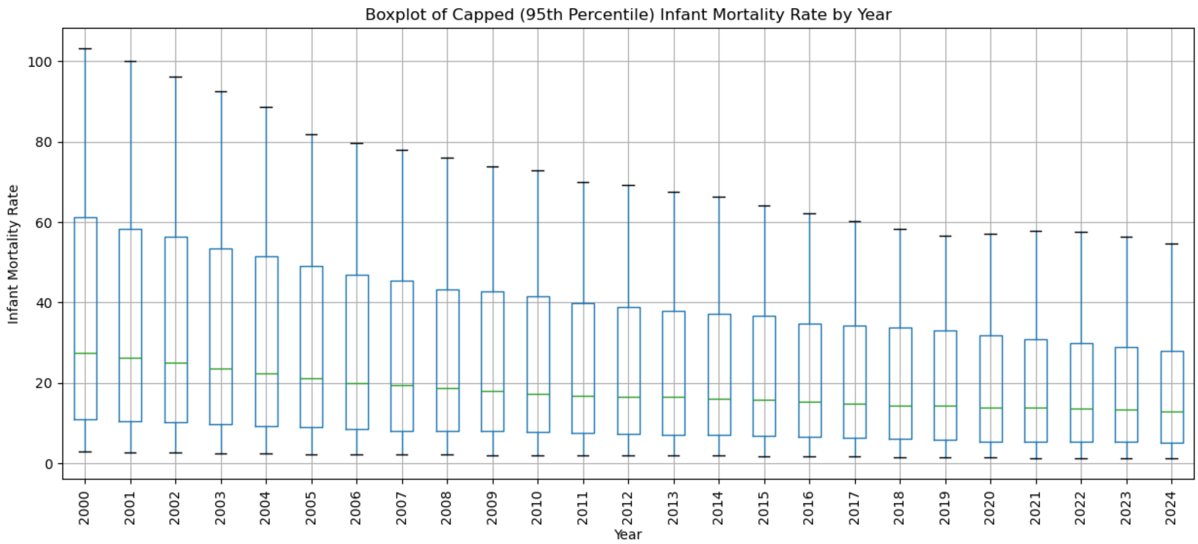

Figure 2. Infant Mortality Rate After 95th Percentile Cap

For the disease_df dataset, higher mortality values were identified in countries such as Afghanistan, Kazakhstan, the Russian Federation and Ukraine. In this case, the distribution remained relatively balanced and the outliers were not excessively far from the rest of the dataset. Since these values reflected realistic non-communicable disease risks and broader healthcare conditions, no capping or removal was applied.

2. SDG 6 Dataset (stress_df)

Within the stress_df dataset, countries such as Bahrain, Kuwait, Libya and Egypt appeared as potential outliers due to extremely high freshwater stress values. These values were considered realistic because the countries experience severe water scarcity and high dependence on freshwater withdrawal. The high values also appeared consistently across multiple years, indicating genuine environmental conditions rather than reporting issues. Because of this, no outlier treatment was applied.

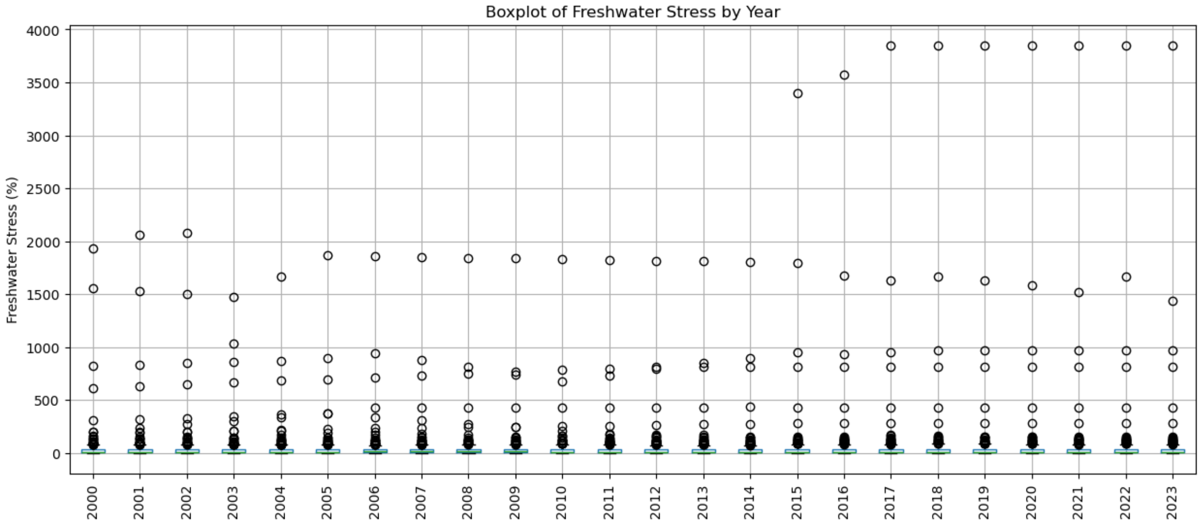

Figure 3. Freshwater Stress Retained Outliers

3. SDG 11 Datasets (disaster_df)

For the disaster_df dataset, extremely high values were retained because disaster events can naturally create major spikes in affected populations during certain years. These values reflected genuine disaster impacts rather than anomalies. However, negative values were replaced with 0 because a negative number of affected people is not logically possible for this type of indicator.

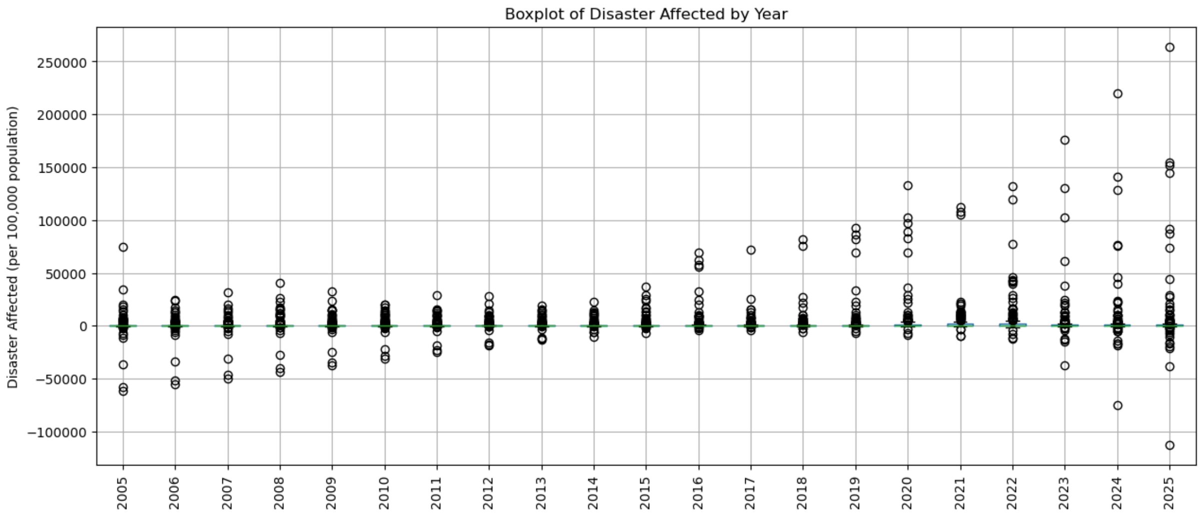

Figure 4. Disaster Affected Negative Outlier Values

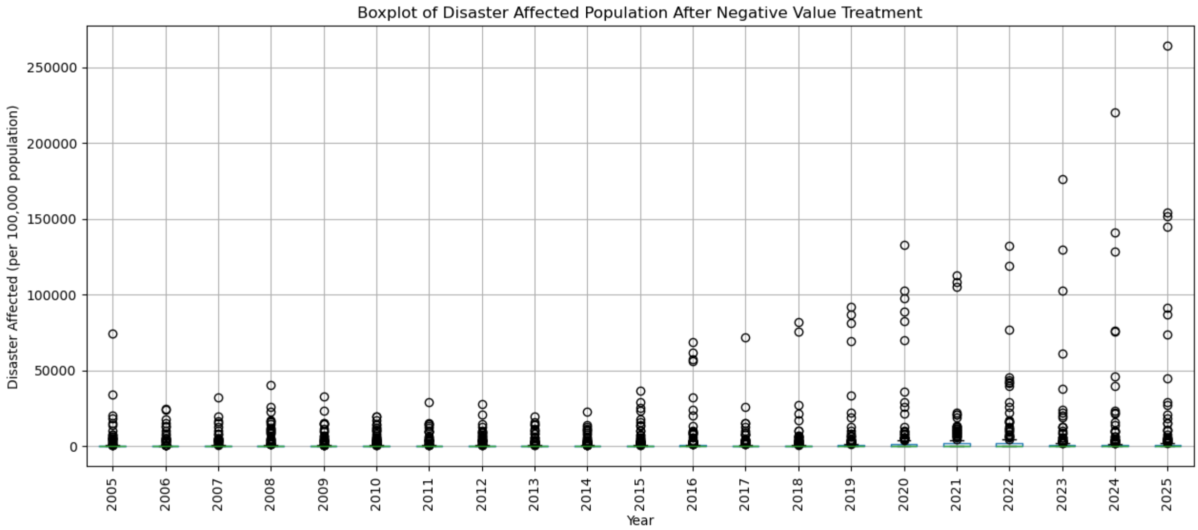

Figure 5. Disaster Affected Retained Positive Outliers

4.5 Variable Creation

To estimate future SDG readiness, projected 2030 variables were created for every indicator using historical trend data. A consistent forecasting structure was applied across all SDG categories to maintain comparability throughout the project. The forecasting years selected were:

- 2005
- 2010
- 2015
- 2020

The decision to use 5-year intervals was mainly influenced by datasets that did not contain complete yearly observations, particularly within SDG 3. Applying the same interval structure across all indicators helped maintain methodological consistency throughout the forecasting process, just as mentioned earlier.

The most recent available year was also included within each dataset to improve the realism of the 2030 projections. A linear regression trend was then calculated individually for each country and this trend was used to estimate projected values for 2030.

Countries with missing values across the selected forecasting years were excluded from prediction to avoid generating unreliable estimates from incomplete trend patterns.

The final forecasting datasets only retained:

- GeoAreaName
- The projected 2030 variable

This simplified structure was intentionally used because the projected variables would later be merged together.

GeoAreaName	InfantMortality_2030	DiseaseMortality_2030	SuicideMortality_2030
Afghanistan	37.457082312474995	28.517286652078724	3.1458424507658833

GeoAreaName	Stress_2030_Predicted	DrinkingWater_2030_Predicted	Handwashing_2030_Predicted
Afghanistan	54.75999999999999	35.76655704585755	56.782833608890314

GeoAreaName	SlumPopulation_2030_Predicted	DisasterAffected_2030_Predicted
Afghanistan	75.33906050259952	1609.3935138856468

4.6 Merging Sub-Indexes

When combining the forecasting datasets into their SDG sub-indexes, an outer join was used instead of an inner join because not every country was present across every indicator dataset.

Using an inner join would have removed countries that were missing just one indicator, which would greatly reduce country coverage across the project. Even if a country only had data for some indicators, it was still considered valuable because it could still provide meaningful evidence of SDG readiness.

For this reason, all countries appearing in at least one dataset were retained within the final SDG 3, SDG 6 and SDG 11 sub-indexes.

After merging, all remaining missing values were filled with 0. This allowed countries with incomplete reporting to still remain in the analysis while reflecting the absence of measurable readiness for the missing indicator. This approach helped preserve country coverage while still ensuring that missing indicators negatively affected the final readiness score.

5. Multivariate Analysis

5.1 Correlation Analysis (Sub-Index Indicators)

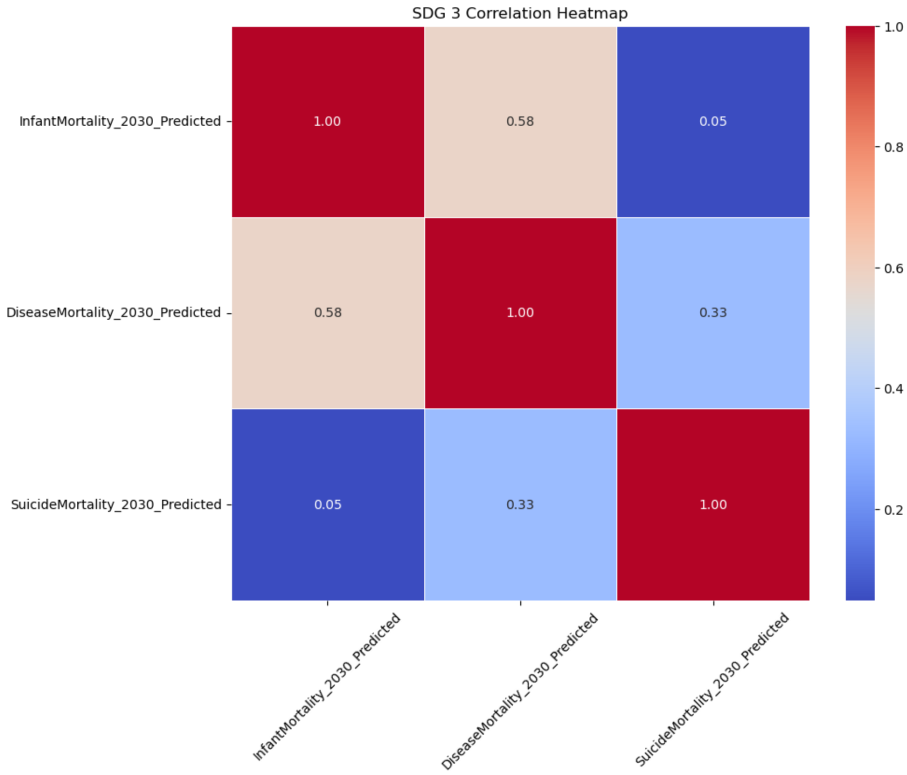

Figure 6. SDG 3 Indicators Multicollinearity

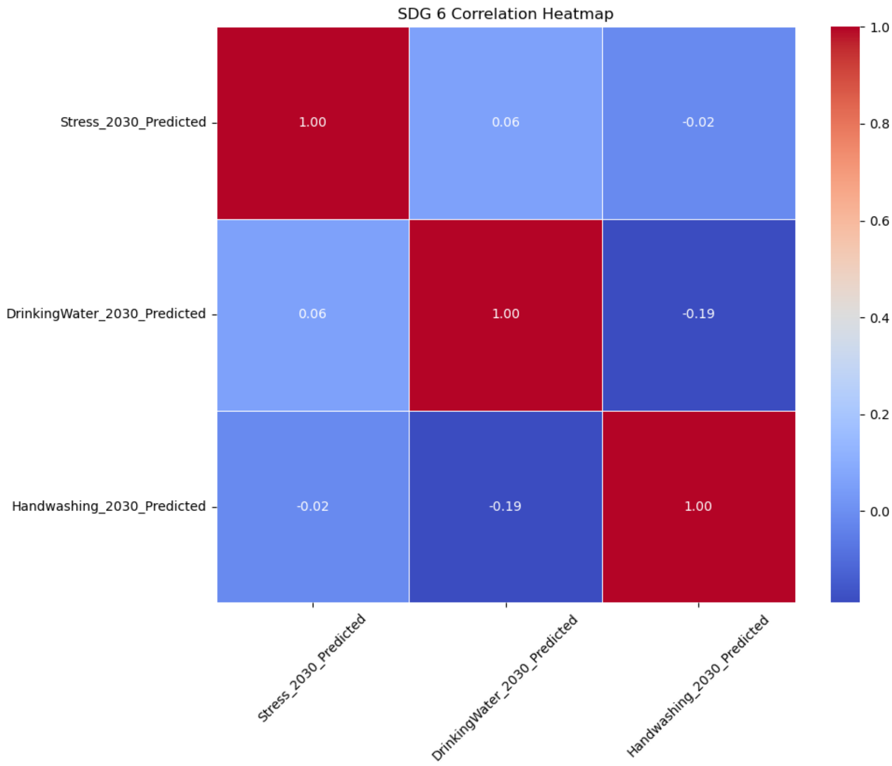

Figure 7. SDG 6 Indicators Multicollinearity

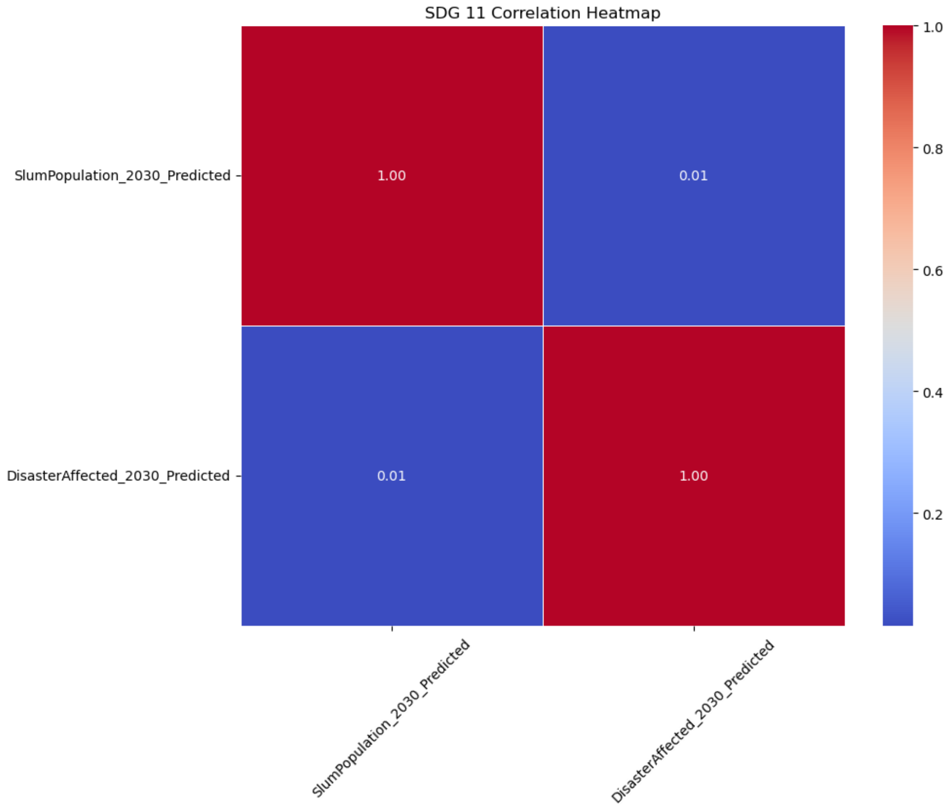

Figure 8. SDG 11 Indicators Multicollinearity

Correlation heatmaps were used as an initial method to inspect the relationships between indicators within each SDG sub-index before the datasets were fully combined.

Across SDG 3, SDG 6 and SDG 11, most correlation values remained low to moderate, with no extremely strong relationships identified between variables. The strongest relationship appeared within the SDG 3 indicators, while SDG 6 and SDG 11 generally showed very weak correlations close to zero.

These results suggested that the indicators were sufficiently independent and were not heavily measuring the same underlying issue across countries. This reduced concerns surrounding any redundancy and multicollinearity during the early stages of sub-index construction.

5.2 Correlation Analysis (Composite Indicators)

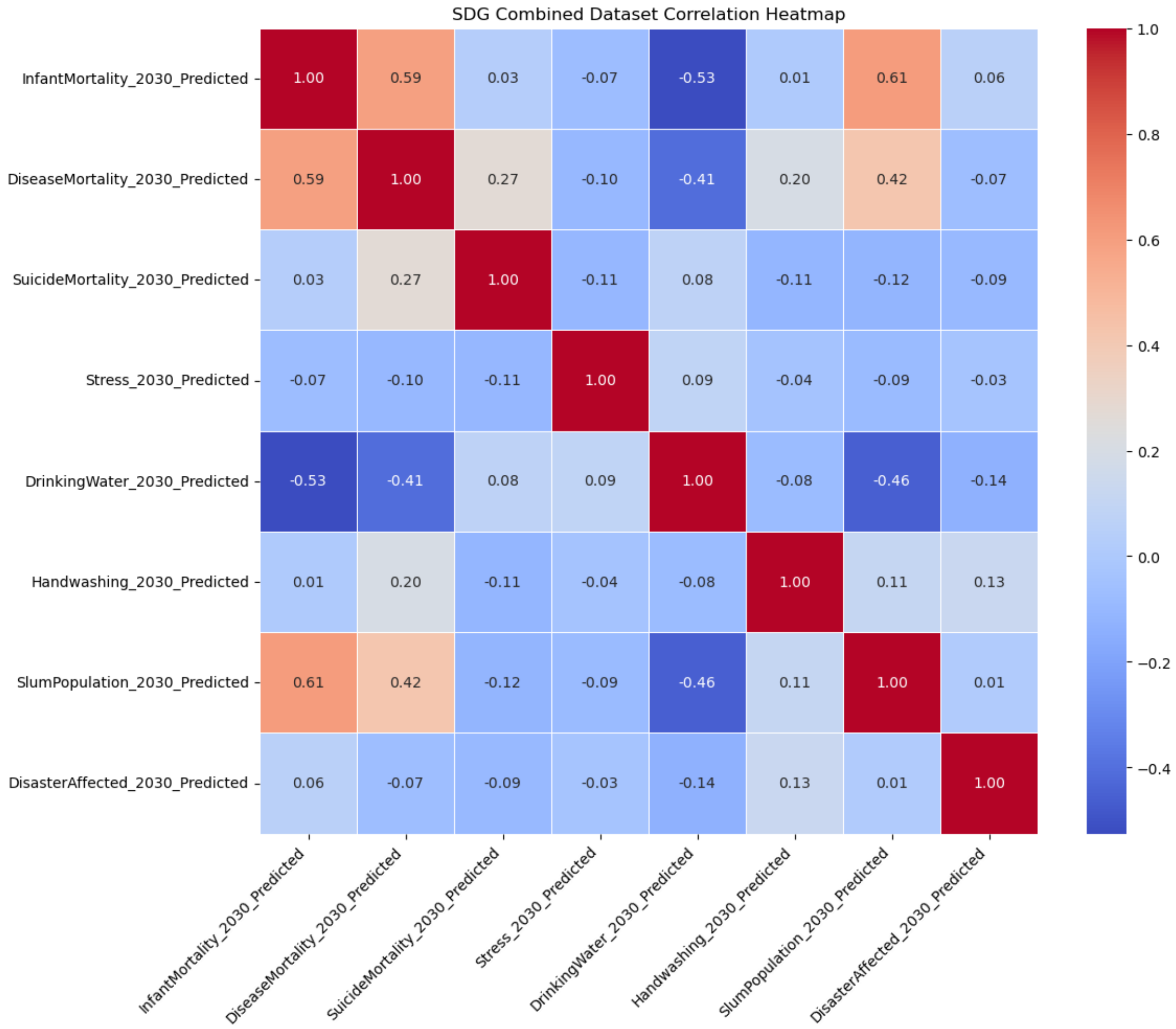

Figure 9. SDG All Indicators Multicollinearity

1. Strong Positive Correlations

- SlumPopulation_2030_Predicted vs. InfantMortality_2030_Predicted (0.61): This represents the strongest positive correlation in the dataset. This correlation is expected, as poorer living conditions and limited healthcare access within slum areas naturally drive higher infant mortality rates. However, at 0.61, the correlation is not high enough to suggest problematic multicollinearity. Both variables capture the distinct socioeconomic factors.
- InfantMortality_2030_Predicted vs. DiseaseMortality_2030_Predicted (0.59): This strong relationship reflects how weaker overall healthcare infrastructure impacts multiple health metrics simultaneously. Despite the overlap, they both measure fundamentally different aspects of public health.
- SlumPopulation_2030_Predicted vs. DiseaseMortality_2030_Predicted (0.42): A moderate and reasonable correlation, highlighting how poor urban living conditions negatively affect long-term community health outcomes.

2. Strong Negative Correlations

- InfantMortality_2030_Predicted vs. DrinkingWater_2030_Predicted (-0.53): This is the strongest negative correlation, which logically shows that better access to safe drinking water directly relates to lower infant mortality rates.
- SlumPopulation_2030_Predicted vs. DrinkingWater_2030_Predicted (-0.46): This expected negative correlation indicates that regions with larger slum populations generally face steeper challenges in providing reliable clean water infrastructure.

- DiseaseMortality_2030_Predicted vs. DrinkingWater_2030_Predicted (-0.41): This underscores how improved clean water access serves as a foundational driver for lowering disease mortality and lifting overall public health.

5.3 Principal Component Analysis

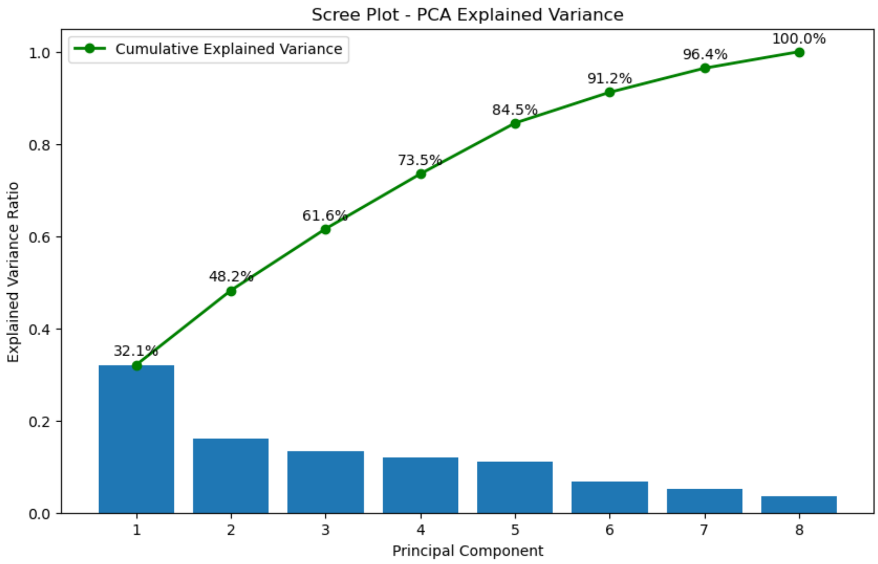

Figure 10. Scree Plot Variance of PCs

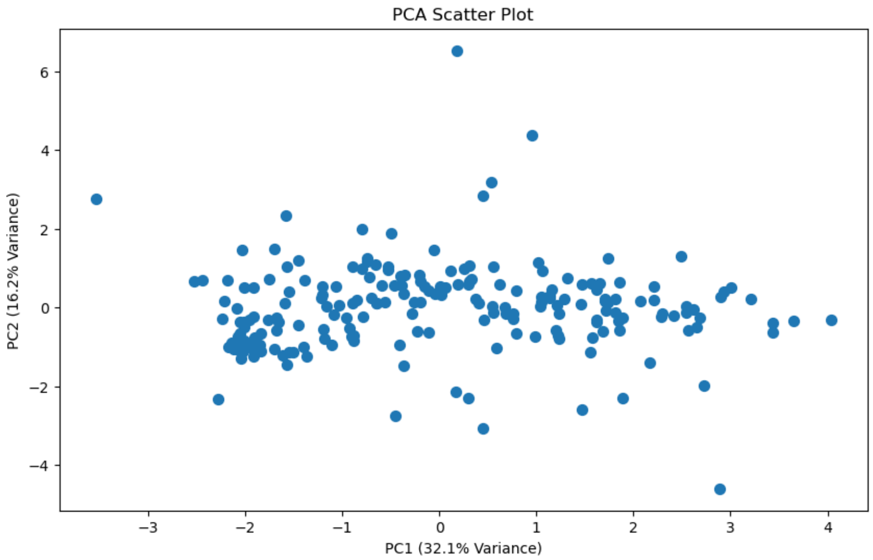

Figure 11. Scatterplot PC1 and PC2 Variance

The PCA results showed a relatively balanced spread of variance across multiple components rather than a single dominant factor controlling the dataset. The scree plot indicated that PC1 explained approximately 32.1% of the total variance, while PC2 explained around 16.2%. Together, the first two principal components accounted for roughly 48.2% of the dataset's total variance.

This suggests that the SDG indicators were not overly dependent on one another and that the dataset captured multiple dimensions of country performance. If one component had explained most of the variance alone, it could indicate excessive overlap or redundancy between the selected indicators.

The cumulative variance reached approximately 73.5% by PC4 and 84.5% by PC5, showing that a relatively small number of principal components preserved most of the information contained within the original dataset. This indicates that the dataset structure remained efficient while still maintaining diversity between variables.

The number of PCs retained using the Kaiser Criterion was 3, which further supported this interpretation, with only the first 3 principal components producing eigenvalues greater than 1. This suggests that these components contained the most meaningful variation within the dataset, while the remaining components contributed comparatively limited additional information.

5.4 K-Means Clustering

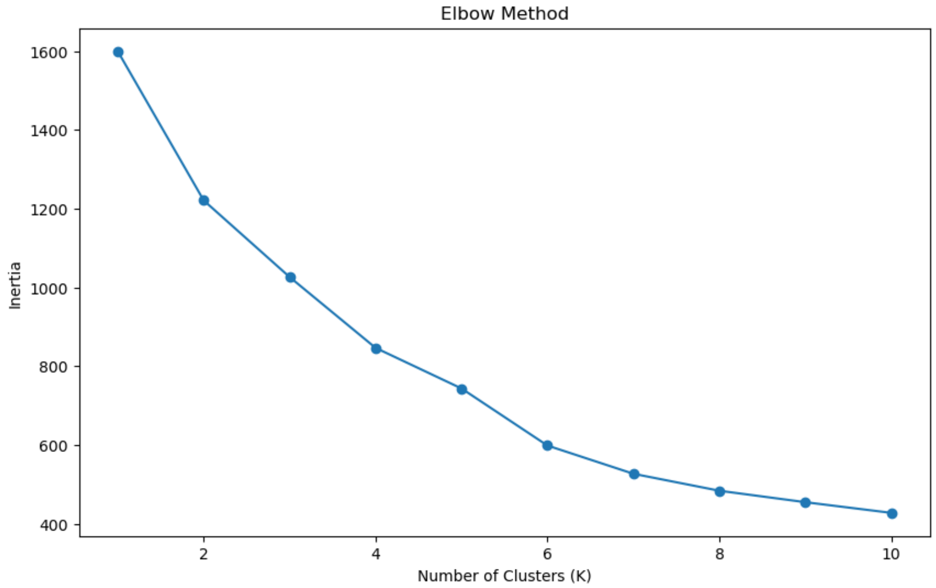

Figure 12. K-Means Clustering Elbow Method

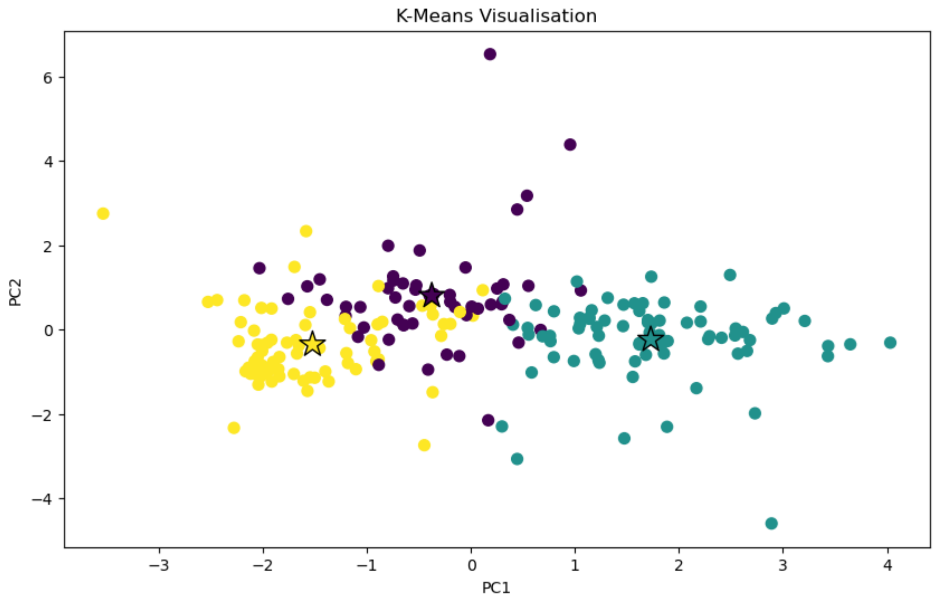

Figure 13. K-Means Cluster Variance

The elbow plot suggested that using approximately 3 clusters was an appropriate choice for the dataset. The inertia value decreased sharply between K=1 and K=3, indicating that additional clusters significantly improved the grouping of countries during the early stages. After K=3, the curve began to flatten more gradually, suggesting diminishing returns from adding further clusters.

Using 3 clusters provided a reasonable balance between simplifying the dataset and still capturing meaningful differences between countries. Too many clusters could reduce interpretability, while too few clusters could oversimplify the variation within the SDG indicators.

The K-Means visualisation showed that the clusters were reasonably separated across the PCA space, particularly along the PC1 axis. This suggests that the dataset contained noticeable differences between groups of countries in terms of overall SDG performance patterns.

The centroids represented the average position of countries within each cluster. The spread of observations around the centroids appeared realistic, since countries naturally vary across different SDG indicators rather than behaving identically.

The turquoise cluster appeared more dispersed, particularly towards the positive PC1 direction, which may indicate greater variability in SDG characteristics among countries within that group. In contrast, the yellow cluster appeared more compact, suggesting more similar development patterns between the countries belonging to that cluster.

6. Normalisation

6.1 Choosing Methodology

After completing the multivariate analysis, including PCA and K-Means clustering, Z-score standardisation was selected as the normalisation method for the dataset.

This approach was chosen because the SDG indicators were measured using very different units and scales, including mortality rates, percentages and disaster-related population counts. Without normalisation, variables with larger numerical ranges could dominate the analysis and distort the final composite indicator.

According to the Organisation for Economic Co-operation and Development Handbook on Constructing Composite Indicators, standardisation is an important step when combining indicators measured on different scales into a single comparable framework.

The handbook specifically identifies *"Z-score standardisation as a method that converts indicators onto a common scale using the mean and standard deviation of the dataset."*

Z-score standardisation was therefore considered appropriate for this project because it allowed all indicators to become directly comparable while still preserving the variation between countries.

6.2 Why Z-Scores Work

According to the paper "Standardization and Weighting of Composite Indicators: The OECD Approach":

"Standardisation consists of transforming the variables so that they have a mean equal to zero and a standard deviation equal to one."

This means:

- Positive values represent countries performing above the dataset average for a particular indicators.
- Negative values represent countries performing below the average.

The transformed results also helped highlight differences in variation across indicators. Variables such as SlumPopulation_2030_Predicted and InfantMortality_2030_Predicted showed larger positive Z-scores for some countries, indicating significantly higher developmental challenges compared to the dataset average.

In contrast, indicators such as Stress_2030_Predicted appeared more tightly distributed around zero, suggesting less variation between countries for that specific measure.

7. Weighting & Aggregation

7.1 Determining Direction

Before applying weighting and aggregation, all indicators were adjusted so that they moved in the same direction. This was an important step because the final SDG readiness index needed to be interpreted consistently, where higher values always represented stronger SDG readiness and lower values represented weaker readiness.

Several indicators naturally behaved as negative indicators, meaning higher values reflected poorer outcomes. These included:

- Infant Mortality
- Disease Mortality
- Suicide Mortality
- Water Stress
- Slum Population
- Disaster-Affected Population

These indicators were reversed after standardisation so that higher transformed values would instead represent stronger SDG performance.

In contrast, indicators such as drinking water access and handwashing access already behaved as positive indicators because higher values naturally reflected stronger sanitation, infrastructure and public health conditions. These variables therefore remained unchanged.

According to the Organisation for Economic Co-operation and Development Handbook on Constructing Composite Indicators:

"All variables should be coded in the same direction so that higher values correspond to better performance."

This helped ensure consistency throughout the final composite indicator framework.

7.2 Equal Weighting

An equal weighting approach was applied across all indicators within each SDG category.

This approach was selected because the earlier multicollinearity analysis showed relatively low correlations between indicators, suggesting that the variables remained sufficiently independent from one another. Since the indicators measured different dimensions of sustainable development, no single variable was considered more important than another.

Equal weighting also helped maintain balance across the SDG framework. Healthcare outcomes, sanitation access, water sustainability, urban living conditions and disaster resilience all represent different development challenges and therefore contributed equally towards the final readiness index.

The weighting structure applied was:

- Equal weighting across SDG 3 indicators (1/3)
- Equal weighting across SDG 6 indicators (1/3)
- Equal weighting across SDG 11 indicators (1/2)

This approach was considered appropriate for maintaining fairness, transparency and interpretability within the project.

7.3 Arithmetic Mean

The Weighted Arithmetic Mean method was used to aggregate indicators into SDG sub-indexes and later into the final overall SDG readiness index.

This method was selected because it is simple to interpret, widely used in composite indicator construction and allows all indicators to contribute proportionally towards the final score.

The arithmetic mean also preserves the contribution of each individual indicator while producing a clear overall readiness measurement across countries.

According to the Organisation for Economic Co-operation and Development Handbook on Constructing Composite Indicators:

"Linear aggregation is the most common aggregation method."

The handbook further explains that arithmetic aggregation assumes full compensability, meaning stronger performance in one indicator can offset weaker performance in another. Within this project, this was considered acceptable because SDG readiness reflects multiple dimensions of development rather than a single isolated issue.

8. Linking to Other Indices

8.1 Existing Indices

Figure 14. UN Division for Sustainable Development Goals

Several existing international frameworks and ranking systems already attempt to measure global development, sustainability and country performance across multiple dimensions. These existing indices helped support the overall concept behind constructing a composite SDG readiness indicator within this project.

The United Nations Sustainable Development Goals framework highlights that the SDGs are highly interconnected rather than functioning as isolated goals. According to the Council on Foreign Relations:

"The SDGs are interconnected."

This idea strongly supports the structure of this project because health, sanitation, water sustainability, urban development and disaster resilience naturally influence one another across countries. For example, poor sanitation can contribute towards higher disease mortality, while weak urban infrastructure can increase vulnerability to disasters and poorer living conditions.

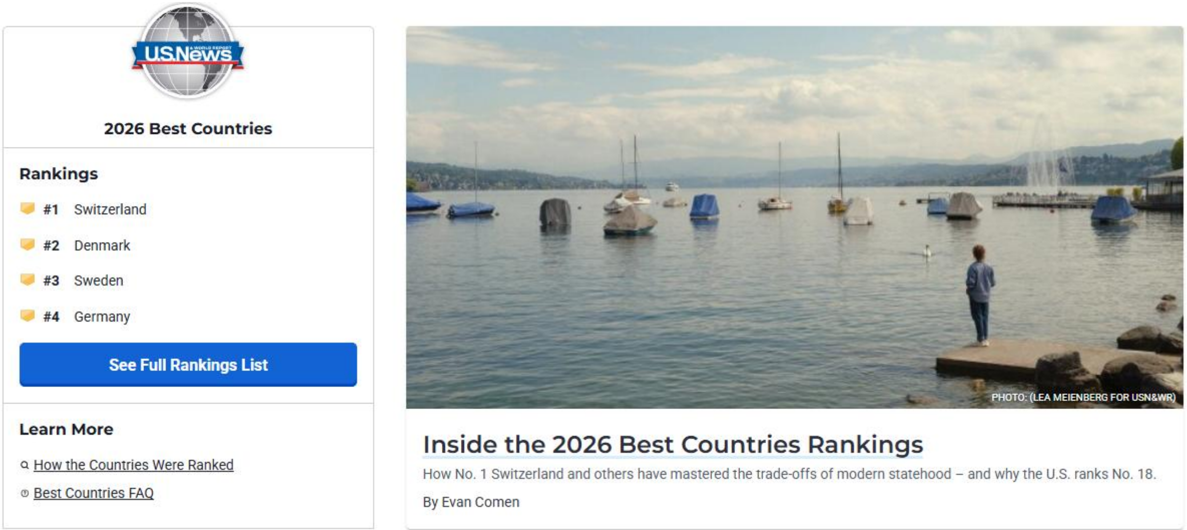

Figure 15.The overall ranking of Best Countries measure global performance

This project also shares similarities with broader international ranking systems such as the U.S. News & World Report Best Countries Rankings. That index evaluates countries using multiple development dimensions including:

- Health
- Governance
- Natural Environment
- Tourism
- Economics
- Opportunity
- Infrastructure

Although the methodology differs, both projects follow a similar principle by combining multiple indicators into a single comparative framework used to evaluate country-level performance. This further supports the idea that composite indicators are useful for simplifying complex development issues into interpretable rankings and comparisons.

8.2 Cross-Indicator Analysis

This project used a cross-indicator approach by combining multiple indicators across SDG 3, SDG 6 and SDG 11 into a single composite readiness framework.

Rather than relying on one variable alone, the project incorporated indicators related to:

- Infant Mortality
- Disease Mortality
- Suicide Rates

- Freshwater Sustainability
- Freshwater Under Stress
- Sanitation Access
- Slum Population
- Disaster Resilience

This approach was considered important because sustainable development cannot realistically be measured using a single dimension. Countries may perform strongly in one area while facing significant challenges in another. Combining multiple indicators therefore created a broader and more balanced representation of national SDG readiness.

The indicators selected within this project were also intentionally chosen because they addressed different but interconnected dimensions of development. SDG 3 focused primarily on health and well-being, SDG 6 addressed water and sanitation infrastructure, while SDG 11 focused on urban sustainability and disaster vulnerability.

By combining these areas together, the project attempted to capture a wider picture of long-term development readiness rather than measuring one isolated issue. This helped justify the use of a composite indicator framework while also supporting the interconnected nature of the SDGs themselves.

9. Visualisation of Results

9.1 Top Performers (SDG-based)

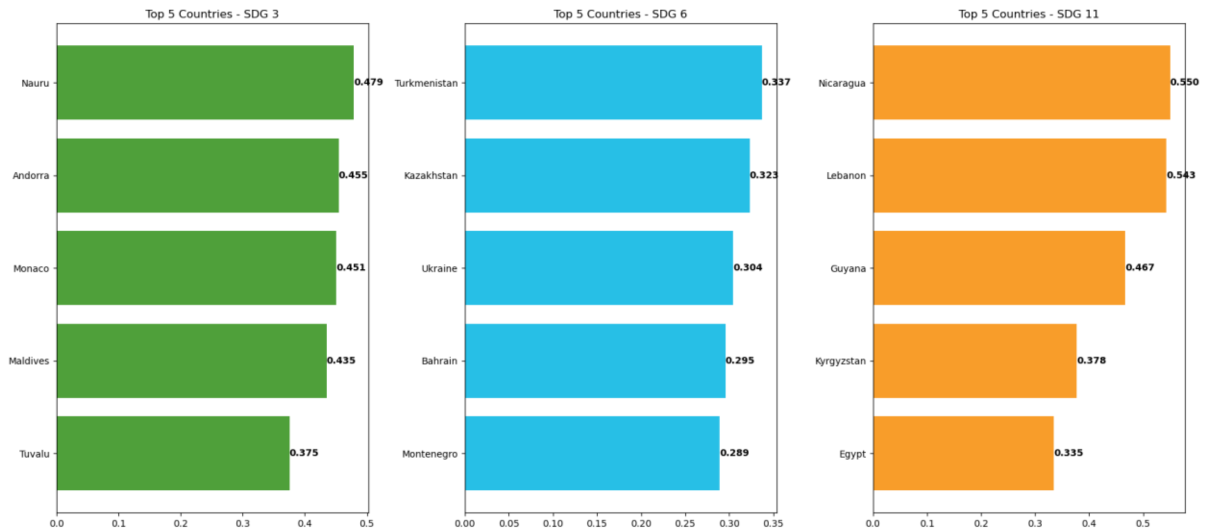

Figure 16. SDG Based Top 5

SDG 3: Health & Well-Being

Nauru, Andorra and Monaco achieved the strongest SDG 3 scores. It is likely that these countries benefit from better healthcare access, stronger medical services and higher overall living standards, which may contribute to lower mortality rates and improved public health outcomes.

Maldives and Tuvalu also performed well. This may suggest that healthcare conditions and public health support within these smaller island nations have gradually improved over time.

SDG 6: Clean Water & Sanitation

The top SDG 6 performers included Turkmenistan, Kazakhstan and Ukraine. It is possible that these countries have more developed water distribution systems and broader sanitation coverage compared to many others in the dataset.

Bahrain and Montenegro also ranked highly, which may reflect relatively stable infrastructure and consistent access to drinking water and sanitation services.

SDG 11: Sustainable Cities & Communities

Nicaragua, Lebanon and Guyana achieved the strongest SDG 11 scores. This may suggest that these countries currently experience lower slum population pressures or more stable disaster-related conditions compared to others included in the analysis.

Kyrgyzstan and Egypt also performed well, possibly reflecting more stable urban development trends and improving resilience toward environmental or infrastructure-related challenges.

9.2 Bottom Performers (SDG-Based)

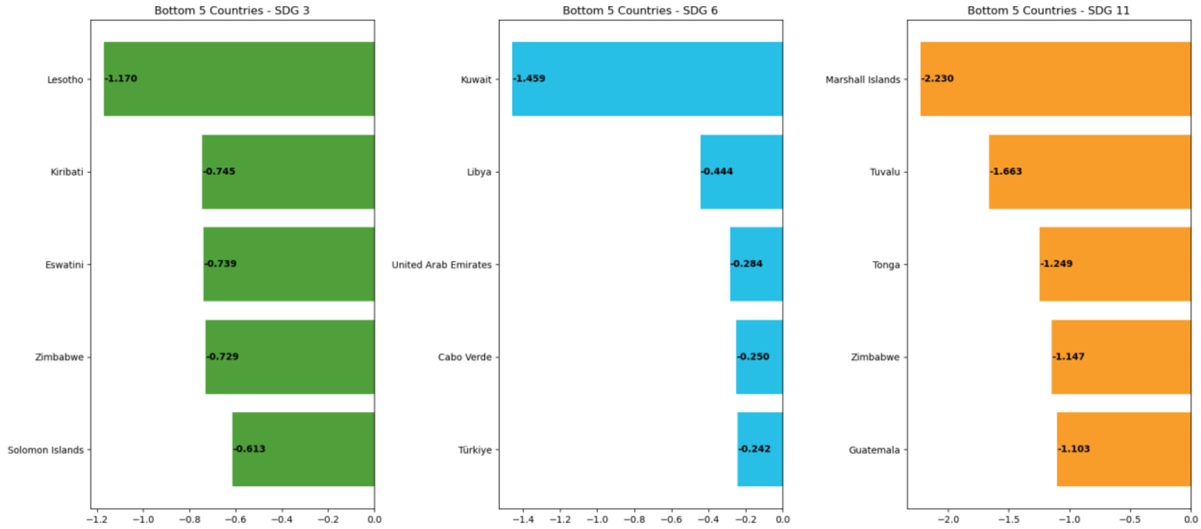

Figure 17. SDG Based Bottom 5

SDG 3: Health & Well-Being

The lowest-performing countries in SDG 3 included Lesotho, Kiribati and Eswatini. It is possible that these countries continue to face challenges related to healthcare access, medical infrastructure and wider public health conditions, which may contribute to weaker health outcomes overall.

Zimbabwe and Solomon Islands also ranked poorly, which may reflect ongoing pressure on healthcare systems and difficulties in improving long-term public health indicators.

SDG 6: Clean Water & Sanitation

The weakest SDG 6 performers were Kuwait, Libya and United Arab Emirates. Their lower rankings may be linked to high water stress levels, environmental limitations or long-term pressure on water sustainability systems.

Cabo Verde and Türkiye also showed lower readiness scores, possibly suggesting growing pressure on sanitation infrastructure and water resource management.

SDG 11: Sustainable Cities & Communities

The lowest-performing countries in SDG 11 included Marshall Islands, Tuvalu and Tonga. These countries are likely more vulnerable to environmental and disaster-related risks, which may strongly affect urban sustainability and community resilience.

Zimbabwe and Guatemala also ranked among the weakest performers, possibly reflecting ongoing urban development and infrastructure-related challenges toward 2030.

9.3 Top 50 Countries

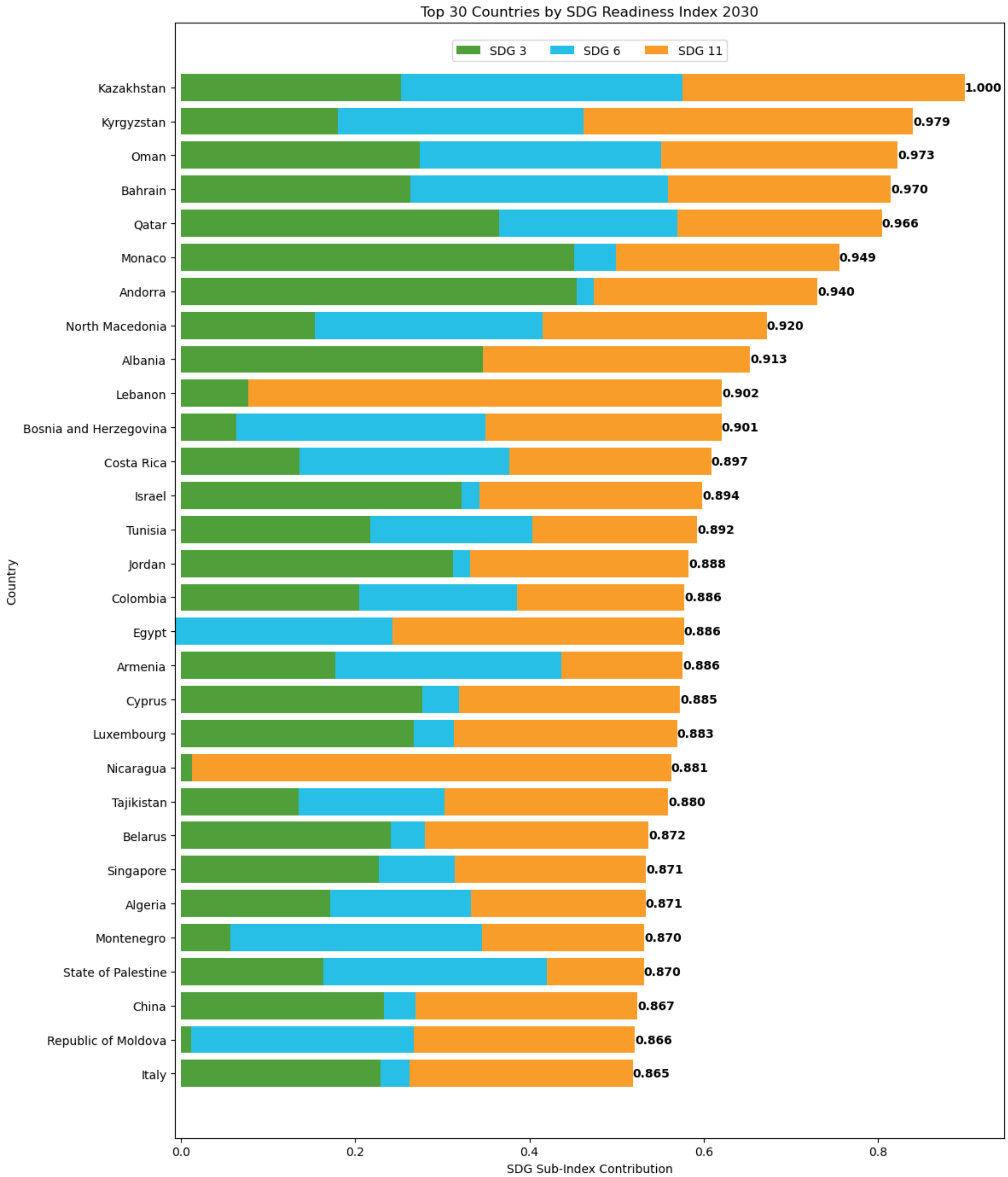

Figure 18. Top 50 Countries Ranking

The overall results show that countries do not need to perform equally across every SDG category to achieve a strong overall SDG Readiness Index by 2030. Many of the top-performing countries displayed uneven sub-index contributions, where one or two SDGs contributed much more strongly than the others.

For example, Kazakhstan and Kyrgyzstan showed particularly strong performance in SDG 6 and SDG 11, while countries such as Monaco and Andorra were heavily driven by stronger SDG 3 performance. Similarly, Lebanon and Nicaragua relied more strongly on SDG 11 contributions.

This suggests that overall SDG readiness is not necessarily dependent on balanced development across all indicators. Instead, countries that perform exceptionally well in certain SDG areas can still achieve high overall readiness scores, even if other SDGs contribute less strongly.

In real-world terms, this may reflect how countries prioritise development differently based on their economic structure, geography, infrastructure, healthcare systems or environmental conditions. Some countries may naturally progress faster in healthcare, while others may show stronger improvements in sanitation, urban sustainability or disaster resilience.

Overall, the results suggest that although SDG progress varies between countries, strong performance in key development areas may still place countries in a favourable position to achieve broader SDG readiness toward 2030.

10. Conclusion

10.1 Summary of Findings

This project successfully developed a composite indicator framework used to measure and compare SDG readiness across countries by 2030. The final Global SDG Readiness Index combined indicators from SDG 3 (Good Health and Well-Being), SDG 6 (Clean Water and Sanitation) and SDG 11 (Sustainable Cities and Communities) into a single comparative framework.

Datasets obtained from the United Nations SDG Data Portal were cleaned, forecasted, normalised and aggregated to create projected 2030 readiness variables for each country. The use of consistent forecasting structures, missing data handling and outlier treatment helped improve the reliability and comparability of the dataset.

The multivariate analysis showed relatively low multicollinearity between indicators, suggesting that the selected variables measured different dimensions of sustainable development rather than heavily overlapping patterns. PCA and K-Means clustering also demonstrated noticeable variation in SDG readiness patterns between countries.

The results showed that indicators related to healthcare, sanitation, urban conditions and environmental vulnerability are interconnected across countries. Countries with stronger healthcare systems, better sanitation access, lower slum populations and lower disaster vulnerability generally achieved stronger readiness scores.

Overall, the project demonstrated how composite indicators can simplify complex SDG data into a more interpretable framework while still preserving meaningful differences between countries.

10.2 Future Improvements

Although the datasets used in this project were obtained from the official United Nations SDG Data Portal and are internationally recognised, some indicators still contained missing values, inconsistent yearly coverage and reporting gaps across countries. Future improvements could involve using more complete and higher-resolution datasets where available to improve the accuracy and reliability of the forecasting process and final readiness scores.

Another improvement would be expanding the project to include additional SDGs and indicators. This project focused on SDG 3, SDG 6, and SDG 11, which provided strong insight into health, sanitation, and sustainability. However, including further SDGs such as education, poverty, climate action, governance or economic development could create a more balanced representation of overall SDG readiness.

At times, some indicators within the current framework may outweigh others during aggregation, which can influence country rankings more heavily in certain areas. Adding additional SDG categories would help distribute the influence of indicators more evenly and reduce the possibility of rankings being overly shaped by a smaller number of variables.

Future work could also explore alternative forecasting techniques, weighting methods and sensitivity analysis to further test the robustness of the composite indicator framework. More advanced machine learning or time-series forecasting approaches may also improve the accuracy of projected 2030 readiness estimates.

10.3 Acknowledgements

The author of this report would like to extend sincere gratitude towards his peers for their continued support, advice and shared suffering throughout the development of this project (sarcasm).

The author would also like to give special thanks to his two cat companions, who remained beside him throughout countless late-night analysis sessions. Although their primary contributions consisted of demanding food at midnight, walking across the keyboard at critical moments and requesting pats during periods of peak concentration, their companionship and moral support were genuinely appreciated.

Finally, the author would like to thank the lecturer for delivering an insightful and engaging module that completely changed the way data analysis and visualisation are viewed. The knowledge and perspective gained throughout the module played a major role in shaping the direction and overall development of this project.

11. Version Control

11.1 GitHub

The entire project was maintained using GitHub for version control and project organisation.

The repository contains all source code, datasets, Jupyter notebooks and outputs required to reproduce the SDG readiness analysis and final composite indicator results.

Project Repository: https://github.com/frogosorap/SDGReadinessIndex.git

11.2 Structure

Raw_Data (Directory):

- Cleaning_SDG_3.ipynb: Notebook responsible for cleaning, preprocessing, forecasting, and preparing all SDG 3 datasets related to health and well-being.
- Cleaning_SDG_6.ipynb: Notebook responsible for cleaning, preprocessing, forecasting, and preparing all SDG 6 datasets related to water and sanitation.
- Cleaning_SDG_11.ipynb: Notebook responsible for cleaning, preprocessing, forecasting, and preparing all SDG 11 datasets related to sustainable cities and disaster vulnerability.
- All 8 raw datasets collected from the United Nations SDG Data Portal
- Combined sub-index datasets created during preprocessing

CA1.ipynb: This notebook contains the main composite indicator workflow. It combines the cleaned SDG sub-index datasets, performs multivariate analysis, applies normalisation, weighting, aggregation and generates the final SDG readiness results and visualisations.

final_ranking.csv: This file contains the final ranked country results after the full composite indicator analysis was completed. The rankings are based on the final aggregated SDG Readiness Index scores generated within the project.

12. Appendix

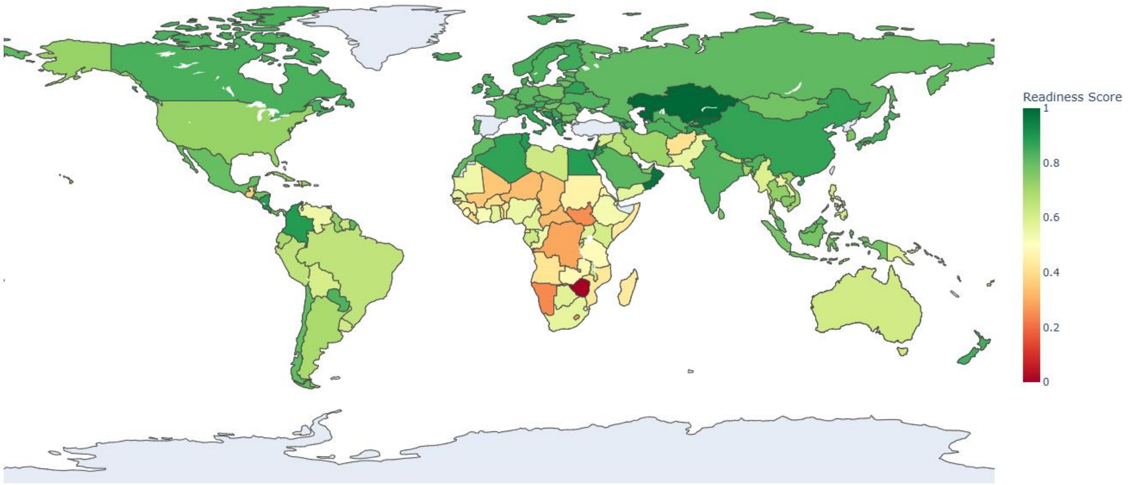

Figure 19. Map of SDG Readiness Index 2030

Rank	Country	Final_SDG_Score
1	Kazakhstan	0.9999999999999999
2	Kyrgyzstan	0.9790143824824078
3	Oman	0.9727488112277008
4	Bahrain	0.9701385427629989
5	Qatar	0.9664505864005142
6	Monaco	0.9491007398721496
7	Andorra	0.9403463082305882
8	North Macedonia	0.9197515333707549
9	Albania	0.913085536
10	Lebanon	0.9016531532344724
11	Bosnia and Herzegovina	0.9013880898819026
12	Costa Rica	0.8972821373674832
13	Israel	0.8937116086634264
14	Tunisia	0.8916364564583835
15	Jordan	0.8880495427379214
16	Colombia	0.8862710113581184
17	Egypt	0.8861246517080879
18	Armenia	0.8856393084386054
19	Cyprus	0.8846062035534514
20	Luxembourg	0.8834338086602699
21	Nicaragua	0.8809809415005126
22	Tajikistan	0.8797274747359869
23	Belarus	0.8717952408182633
24	Singapore	0.8707504729498101
25	Algeria	0.870579721
26	Montenegro	0.8701584005149587
27	State of Palestine	0.8699795270918707
28	China	0.8672376903564669

29	Republic of Moldova	0.8662948695442485
30	Italy	0.8654687567919845
31	Azerbaijan	0.8647652276408827
32	Malta	0.8597923841283381
33	Serbia	0.8593181696399959
34	Greece	0.8533959773506381
35	Finland	0.8516994251333794
36	Denmark	0.8511772558672694
37	Georgia	0.846842667
38	Austria	0.8449763186282493
39	Ireland	0.8441219001263909
40	New Zealand	0.8430996077889957
41	Norway	0.8401197759592175
42	Slovakia	0.8388542747096699
43	Iceland	0.8384958307292681
44	United Kingdom of Great Britain and Northern Ireland	0.837645281
45	Northern Africa and Western Asia	0.8374601409215423
46	Japan	0.8370458710231228
47	Canada	0.8352762605140496
48	Sweden	0.8352036128730049
49	Eastern and South-Eastern Asia	0.8344323078412934
50	Turkmenistan	0.8325203646571481
51	Netherlands (Kingdom of the)	0.8316909483660332
52	Estonia	0.8311860731002256
53	Paraguay	0.8296228806795533
54	Maldives	0.8286784805219343
55	Switzerland	0.8284284791873877
56	India	0.8277589772756064
57	Ukraine	0.8271969354611939
58	Grenada	0.8271848620492857
59	Uzbekistan	0.8271242273863608
60	Czechia	0.8259402478213769
61	Portugal	0.8216751198299629
62	Slovenia	0.8177237077534267
63	Germany	0.8160417275910437
64	World	0.815883526
65	El Salvador	0.8155727074716121
66	Saudi Arabia	0.8141413382777865
67	Belgium	0.8134550572019201
68	France	0.8132837722110275
69	Saint Kitts and Nevis	0.8115133106012944
70	Europe and Northern America	0.8088557263274555
71	Malaysia	0.807608567

72	Russian Federation	0.8068001681174329
73	Latvia	0.8046219973071127
74	Chile	0.8039677900056418
75	Morocco	0.8023818372424026
76	Bhutan	0.7986801414979203
77	Mexico	0.7974514224817529
78	Saint Lucia	0.7956543546674801
79	Lithuania	0.7914250717783996
80	Croatia	0.790521599
81	Romania	0.7860507017425473
82	Bulgaria	0.7858833688776964
83	Belize	0.7836463938292466
84	Mongolia	0.7795455212139656
85	Hungary	0.7783624370764833
86	Poland	0.777364263
87	Honduras	0.7757963543508691
88	Indonesia	0.7754444530178914
89	Republic of Korea	0.7645062551980587
90	Sri Lanka	0.761413675
91	Antigua and Barbuda	0.7611936066754597
92	Barbados	0.7599194827879271
93	Panama	0.7525631301857565
94	Latin America and the Caribbean	0.7524018134196288
95	United Arab Emirates	0.7516903854049581
96	Cambodia	0.7503570923752532
97	Saint Vincent and the Grenadines	0.7459570665844786
98	Dominica	0.7412679325206288
99	Palau	0.7339316669955475
100	Australia and New Zealand	0.7309612932313303
101	Thailand	0.7306233940292107
102	Australia	0.7300407819187246
103	Guyana	0.7288780943339583
104	TÃ¼rkiye	0.7247460740244921
105	United States of America	0.7219244349718815
106	Dominican Republic	0.7218851809187443
107	Iran (Islamic Republic of)	0.7147959253310955
108	Cuba	0.7142048892413158
109	Viet Nam	0.7113071572930092
110	Bangladesh	0.7044332613871003
111	Small island developing States (SIDS)	0.6997242998708223
112	Jamaica	0.6989157155070479
113	Argentina	0.6925065481361763
114	Ecuador	0.6902462506201293
115	Brunei Darussalam	0.6776184389807732

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124	Suriname	0.625597362
125	Libya	0.6253996371761997
126	Fiji	0.6235476164149628
127	Gabon	0.6201323195000581
128	Uruguay	0.6178545800901891
129	Malawi	0.6174444740684161
130	Oceania (exc. Australia and New Zealand)	0.6154452515439331
131	Rwanda	0.6102732394230167
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155	Congo	0.5437195906529904
156	Mauritania	0.543243898
157	Ethiopia	0.5286156573549586
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159	Mauritius	0.5169636487109437

160	CÃ´te d'Ivoire	0.5142762288787013
161	Least Developed Countries (LDCs)	0.5140269868271966
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163	Kiribati	0.4981657207539005
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197	Tuvalu	0.24892747028662715
198	Namibia	0.24341477084681867
199	Marshall Islands	0.004324402
200	Zimbabwe	0

13. References

Research and Databases

Education CFR (2026) The Sustainable Development Goals. Available at: https://education.cfr.org/learn/reading/sustainable-development-goals?gad_source=1&gad_campaignid=23819513442&gbraid=0AAAAAD-E15lKvHzzMc9GZMX5J6t_cykPY&gclid=CjwKCAjwq6DQBhBVEiwA4ZD5XFVKwoVL5WlE57_DJqYQf8N-QnlZjutrvs1z1VhbqWvRrlzjKLMKdhoCMngQAvD_BwE

EUStat (2009) Handbook on constructing composite indicators. Available at: https://www.eustat.eus/prodserv/datos/sem50_i.pdf

OECD (2020) Handbook on Constructing Composite Indicators METHODOLOGY AND USER GUIDE. Available at: https://2526-moodle.dkit.ie/pluginfile.php/1427034/mod_resource/content/1/HandbooktoCreateanIndex1.pdf

SDG Transformation Centre (2026). The Online Library. Available at: https://sdgtransformationcenter.org/online-library

United Nations (2023) SDG Global Database gives you access to data on more than 210 SDG indicators for countries, areas or regions across the globe. Available at: https://unstats.un.org/sdgs/dataportal/

United Nations DESA (2026) The SDG Goals. Available at: https://sdgs.un.org/goals

United Nations Statistics (2023) United Nations Statistics Division SDG API. Available at: https://unstats.un.org/SDGAPI/swagger

US News (2026) 2026 Best Countries. Available at: https://www.usnews.com/news/best-countries

Generative AI Tools

ChatGPT (2026) Generating index cover image. Available at: https://chatgpt.com/

Gemini (2026) Code debugging and assisting with plot charts. Available at: https://gemini.google.com/app

Grammarly (2026) Grammar and spelling errors. Available at: https://app.grammarly.com/